(a) Sources

Multi-Source Interaction

Human Video

Real Robots

Simulation

Align Time & Views

(b) Interaction Schema

SAM2 — Instance masks

DA3 — Metric depth

VLM Caption generation

Actions · states · interactions

Shared interaction schema

Objects	Attributes · states · relations
Actions	Actors · state changes
Evolution	Identity · state history
Context	Goal · time anchors · source

(c) QA Construction

Task rules: Scope · candidates

Record: Plate in rack · t_0

Rule-based Engine

$t_0 \rightarrow t_1 \rightarrow t_2$ | t_3

Visible history | Held out

1. Select evidence & inputs
2. Limit observation scope
3. Build candidates & answer
4. Render via task templates

✓ ***Verify***: Evidence Support

(d) EmboScope

3 Levels · 14 tasks

L1 *4 tasks*
Perception
e.g. Attribute matching

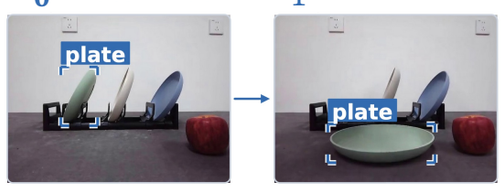
L2 *6 tasks*
Embodied Semantics
e.g. Action between frames

L3 *4 tasks*
Planning & Reasoning
e.g. Action prediction/selection

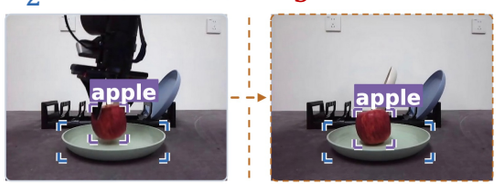
Worked example · one real-robot episode, three levels of questions.

1 Annotated episode

t_0: Plate in rack — t_1: Plate on counter



t_2: Decision cutoff — t_3: Held out



2 Shared records

Goal: apple on green plate

t_0 Concurrent states

Green plate tilted in rack — Red apple resting on counter

$t_0 \rightarrow t_1$ State change

gripper: Grasp → move → release

Plate: rack → counter — tilted → flat

Object evolution

apple: $t_0 \rightarrow t_1$ On counter → t_2 Held above plate

3 Three task levels

L1 Perception — $t_0 \rightarrow t_2$

Q: Which frame shows the plate in the rack?

A: t_0 — state → frame

L2 Embodied semantics — $t_0 \rightarrow t_1$

Q: What changes from t_0 to t_1?

A: Move plate → counter

L3 Planning & reasoning — $goal + \leq t_2$

Q: What action follows this state?

A: Release onto plate